CS 135 — L14: General Recursion

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Counting Up

Count Up Template

```
(define (count-up-template n limit)
  [cond
    [(>= n limit) ...]
    [else (... (count-up-template (add1 n) limit))]])
```

Mutual Recursion

```
(define (keep-next lst)
  (cond
    [(empty? lst) empty]
    [else (cons (first lst) (skip-next (rest lst)))]))

(define (skip-next lst)
  (cond
    [(empty? lst) empty]
    [else (keep-next (rest lst))]))
```

Merge Sort

Merge Sort has the time complexity of $O(n \log n)$

```
;; merge: (listof Num) (listof Num) -> (listof Num)
;; Requires: lists must be sorted in increasing order
(define (merge lst1 lst2)
  (cond
    [(empty? lst1) lst2]
    [(empty? lst2) lst1]
    [(< (first lst1) (first lst2)) (cons (first lst1) (merge (rest lst1) lst2))]
    [(> (first lst1) (first lst2)) (cons (first lst2) (merge lst1 (rest lst2)))]
    [else (cons (first lst1) (cons (first lst2) (merge (rest lst1) (rest lst2))))]))

(define (mergesort lst)
  (cond
    [(or (empty? lst) (empty? (rest lst))) lst]
    [else (merge (mergesort (keep-next lst)) (mergesort (skip-next lst)))]))
```